Public Economics

Lecture 2 · Theoretical Tools of Public Finance

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Today's Questions

- How do individuals choose how much to consume and how hard to work?
- How do firms choose how much to produce?
- What is the theoretical effect of raising — or cutting — Austria's minimum-income support (Sozialhilfe) on economic efficiency?

Based on
Gruber, J. *Public Finance and Public Policy*, Ch. 2; Stantcheva, S., "Theoretical Tools for Public Economics" (Harvard, EC2450A).

Outline

Theoretical vs. Empirical Tools

Theoretical tools
The set of tools designed to understand the *mechanics* behind economic decision making — graphical and mathematical models of utility maximization subject to constraints.

Empirical tools
The set of tools designed to *analyze data* and answer the questions raised by theoretical analysis (the subject of Lecture 3).

Theory tells us the *direction* of an effect; data tells us its *size*. We need both.

2.1 Constrained Utility Maximization

Utility Functions

Utility function
A mathematical mapping of an individual's consumption choices into a level of well-being: $U = u(X_1, X_2, \ldots)$.

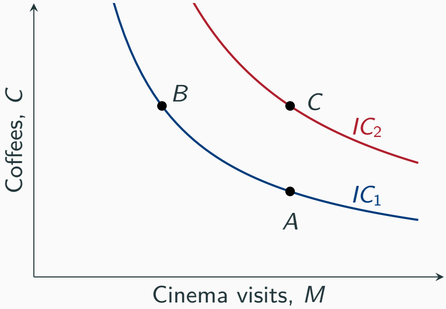
- Consumers are assumed to undertake **constrained utility maximization**: maximize well-being subject to available resources.
- Key assumption: **non-satiation** ("more is better") — more of a good is always at least as good as less.

Running example: Lena, a WU student, choosing between *coffees at the Kaffeehaus* (C) and *cinema visits* (M).

Indifference curve
All bundles of goods that leave an individual equally well off.

- Bundle A: 2 coffees, 1 cinema visit
- Bundle B: 1 coffee, 2 cinema visits
- Bundle C: 2 coffees, 2 cinema visits

Lena is indifferent between A and B, but strictly prefers C to both (more of everything).



Two properties (both from "more is better"): (1) higher indifference curves are preferred; (2) indifference curves always slope downward.

Marginal utility
The additional utility from consuming one more unit of a good: $MU_1 = \partial u / \partial X_1$.

- Example: $U = \sqrt{C \times M}$ (holding $C = 2$ fixed). Moving from 0 to 1 cinema visit raises U from 0 to $\sqrt{2} = 1.41$; from 1 to 2 visits, U rises only to $\sqrt{4} = 2$ (a gain of 0.59); from 2 to 3, the gain is only 0.45.
- This is **diminishing marginal utility**: each additional unit of a good raises utility by less than the previous unit.

Aside: Does Marginal Utility Always Diminish?

Not for addictive goods

Becker & Murphy (1988), "A Rational Theory of Addiction": today's utility from consuming a good can depend on *past* consumption of it — you build a habit.

- The more you have already consumed, the more you enjoy (or "need") consuming more — the opposite of diminishing MU.
- This is a leading theoretical framework for taxing tobacco, alcohol, and sugar: if consumers systematically under-weigh future harm, corrective taxes can improve welfare beyond just correcting externalities.

Marginal Rate of Substitution

Marginal rate of substitution (MRS)
The rate at which a consumer is willing to trade one good for another; equals (minus) the slope of the indifference curve: $MRS_{C,M} = MU_M / MU_C$.

- For $U = \sqrt{C \times M}$: $MRS_{C,M} = C/M$.
- **Diminishing MRS**: as Lena has more cinema visits and fewer coffees, she is willing to give up less and less coffee for one more visit.

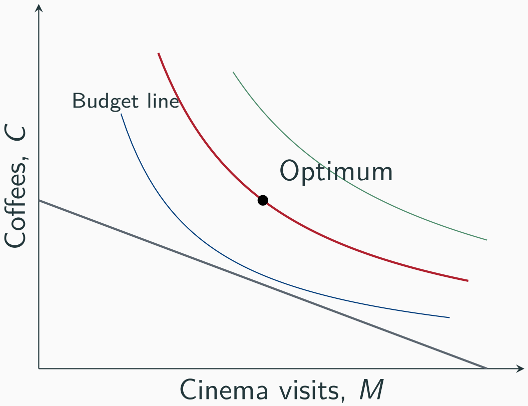
Diminishing MRS is exactly why indifference curves are *convex* (bowed toward the origin).

Budget constraint
All combinations of goods an individual can afford, spending her entire income:
$P_C\, C + P_M\, M = Y$.

- Suppose a Kaffeehaus coffee costs €4, a cinema ticket costs €8, and Lena's monthly "fun budget" is €96.
- Maximum coffees (0 cinema): $96/4 = 24$. Maximum cinema visits (0 coffee): $96/8 = 12$.
- Slope of the budget line: $-P_M/P_C = -8/4 = -2$ — each extra cinema visit costs 2 coffees.

Constrained Choice: The Tangency Condition

- Lena's optimal bundle is where her highest reachable indifference curve is **tangent** to her budget line.
- At that point: $MRS_{C,M} = P_M/P_C$.
- With $U = \sqrt{C \times M}$ and this budget: optimum is $M = 6$, $C = 12$ (she splits her €96 evenly: €48 on each good).



Income and Substitution Effects

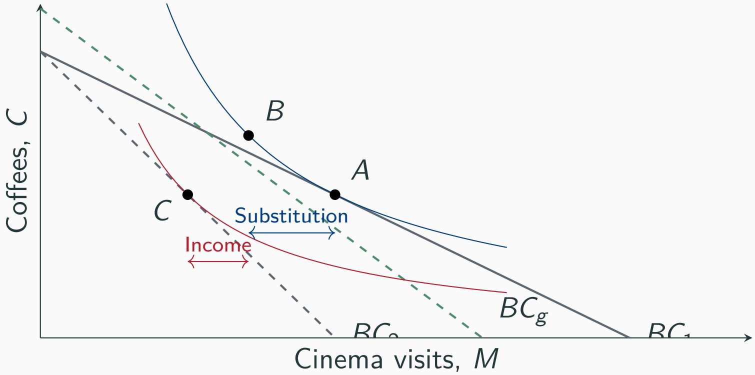
Suppose the price of cinema tickets rises. This has two distinct effects on demand:

Substitution effect
Holding utility constant, a relative price rise *always* causes less consumption of that good.

Income effect
A price rise makes the consumer effectively poorer, typically reducing consumption of *all* normal goods.

For a normal good, both effects push demand in the *same* direction — toward less of the good whose price rose.

Decomposing a Price Change



Cinema price rises: $A \to C$ overall. $A \to B$ (holding utility at IC_1) isolates the *substitution effect*; $B \to C$ isolates the *income effect*.

Normal vs. Inferior Goods

Normal good
Demand *increases* with income (most goods).

Inferior good
Demand *decreases* with income.

Austrian example
Public transport (*Öffis*) use tends to fall as household income rises — once people can afford a car, many drive instead. Public transport behaves like an **inferior good** over part of the income range, even though most goods are normal.

This is exactly why the KlimaTicket (Lecture 1) can only partly be understood through the price effect alone — income effects and preferences over cars vs. transit matter too.

2.2 Putting the Tools to Work: Sozialhilfe and Labor Supply

Austria's Sozialhilfe (Minimum Income Support)

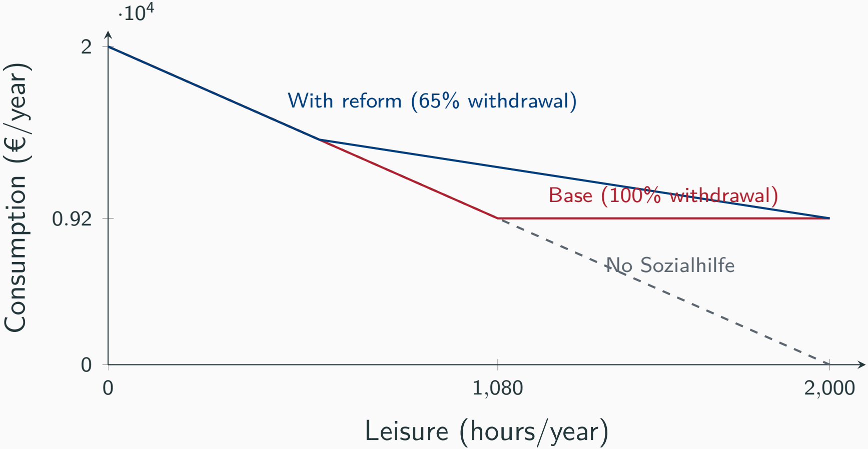
- Since the 2019 *Sozialhilfe-Grundsatzgesetz*, Austria's means-tested minimum income sets **maximum** rates (*Höchstsätze*), not guaranteed minimums, with details varying by *Land*.
- Maximum monthly rate for a single adult/single parent (2024): ≈€1,156. Average *actual* payment to single adults: ≈€766/month.
- § 7 Abs. 6 Sozialhilfe-Grundsatzgesetz (an earnings disregard, commonly called the *Wiedereinsteigerfreibetrag* in secondary sources): up to 35% of net monthly earnings are disregarded for up to 12 months when a recipient starts working.

This is Austria's own version of Gruber's TANF example — a benefit guarantee plus a benefit-reduction rate. Source: Sozialhilfe-Grundsatzgesetz, BGBl. I Nr. 41/2019, § 7 Abs. 6 (RIS: `ris.bka.gv.at/eli/bgbl/i/2019/41/P7`); ÖGB, "So viel Sozialhilfe bekommt man in Österreich."

The Consumption–Leisure Trade-off

- Suppose Julia, a single parent, can work up to 2,000 hours/year at a wage of €10/hour, with no other income and no Sozialhilfe.
- Each hour of leisure "costs" her €10 in forgone wages (opportunity cost).
- Her budget constraint: maximum €20,000/year in consumption (0 leisure) or 2,000 hours of leisure (€0 consumption); slope $= -10$.

Adding Sozialhilfe to the Budget Constraint



Without any earnings disregard, Julia's net return to work is €0 until she earns her way off Sozialhilfe entirely — a pure "welfare trap."

The Wiedereinsteigerfreibetrag: A Real Reduction-Rate Experiment

- **Base regime**: 100% benefit-reduction rate. Every euro earned reduces Sozialhilfe by one euro — net return to work is €0 (flat budget segment).
- **Reform regime**: 35% of net earnings disregarded $\Rightarrow$ effective reduction rate $\approx$65%. Net return to work rises to €3.50 per hour (slope -3.5 instead of flat).

Why this matters

This is *exactly* the benefit-guarantee/reduction-rate logic from Gruber's TANF example — applied to a real 2019 Austrian reform explicitly designed to soften the work disincentive.

Predicted Effects on Labor Supply

- Cutting the reduction rate (100% $\rightarrow$ 65%) makes work pay more — a pure **substitution effect** toward less leisure (more work).
- But it also raises Julia's income at any given hours choice — a competing **income effect** toward *more* leisure.
- The net effect depends on the relative strength of the two — and on how much Julia values leisure relative to consumption (her preferences).

Theory alone cannot say whether labor supply rises or by how much — exactly as in Gruber's Naomi/Sarah comparison.

How Large Will the Response Be?

- The constrained-choice model tells us the *direction* of well-behaved responses — but not their *magnitude*.
- Whether cutting Sozialhilfe's effective reduction rate actually raises single parents' labor supply, and by how much, is an empirical question.

This is exactly what Lecture 3 (empirical tools) is built to answer.

2.3 Equilibrium and Social Welfare

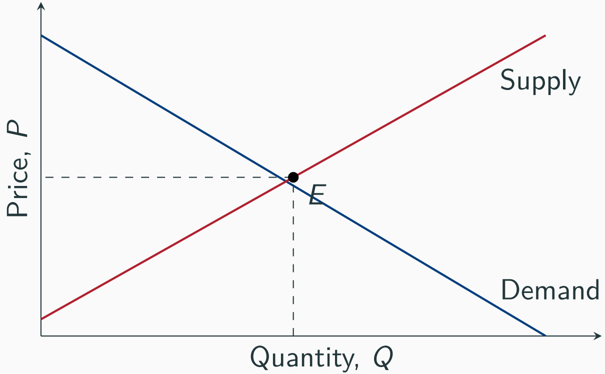
From Individual Choice to Market Demand

- Each individual's utility maximization generates her own demand $X(P, Y)$ for a good.
- The **market demand curve** is the horizontal sum of all individual demands at each price.
- **Elasticity of demand**: $\varepsilon_D = \dfrac{\%\Delta\text{quantity demanded}}{\%\Delta\text{price}}$ — typically negative, and typically not constant along the curve.

Supply Curves and Firms

- Firms use a **production function** to turn inputs (labor, capital) into output, subject to **diminishing marginal productivity**.
- Profit maximization ($\max_Q [\, p \cdot Q - c(Q) \,]$) implies firms produce until **marginal cost equals price**.
- The marginal cost curve *is* the firm's supply curve; market supply is the horizontal sum across firms.

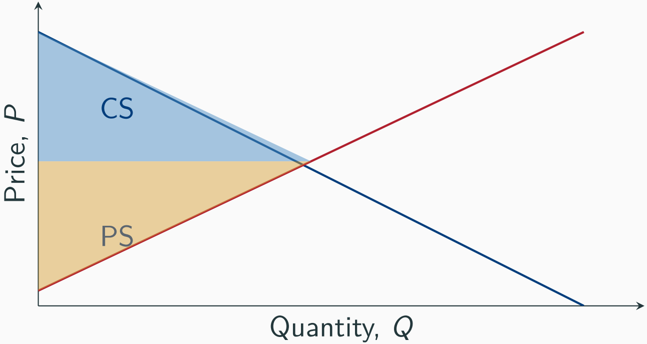
Market Equilibrium



Market equilibrium: the unique price/quantity pair at which demand equals supply, and both sides of the market are satisfied.

Consumer surplus
The value consumers get from a good, above and beyond what they paid. Area below demand, above P_E.

Producer surplus
The profit producers earn, above and beyond their cost. Area above supply, below P_E.



The First Fundamental Theorem of Welfare Economics

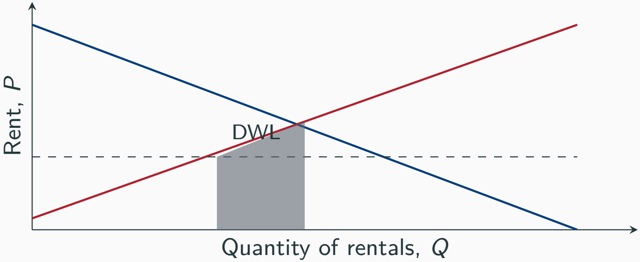
First Fundamental Theorem

The competitive equilibrium, where supply equals demand, maximizes total economic surplus (efficiency).

- Total surplus just counts € — it is blind to *who* gets them.
- Holds under: no externalities, perfect competition, perfect information, rational agents.
- When these assumptions fail (market failures), government intervention can potentially raise — not just redistribute — total surplus.

Application: Rent Regulation in Vienna

- About **21%** of Vienna's rented apartments are subject to the *Richtwertmietzins* price ceiling (currently €6.74/m^2), below the market rent.
- A price ceiling below P_E reduces the quantity landlords are willing to supply — the classic price-ceiling case.



Source: Arbeiterkammer Wien, "Richtwertmietzins"; Stadt Wien, Mietenrechner.

The Second Fundamental Theorem of Welfare Economics

Second Fundamental Theorem
Any Pareto-efficient allocation can be reached by (1) suitably redistributing initial endowments via **lump-sum** transfers, then (2) letting markets work freely.

If this worked in practice, there would be *no conflict* between efficiency and equity.

Why the Second Theorem Fails in Practice

- Lump-sum redistribution requires the government to observe individual *characteristics* (need, ability) — not just *outcomes* (income, work status).
- In reality, the government usually cannot tell these apart, and must base taxes/transfers on observable behavior instead.

Illustration

Suppose 50% of the population is disabled and cannot work (income €0), 50% can work and earn €100. If the government could identify disability directly, it could tax the able €50 *regardless of whether they work*, and give €50 to each disabled person — full redistribution, zero distortion.

But if the government cannot tell a disabled person from an able person who simply isn't working, that same €50 tax-and-transfer destroys the incentive to work — redistribution now costs efficiency.

Social Welfare Functions

Social welfare function (SWF)
A function combining all individuals' utilities into one overall measure of social welfare.

Utilitarian

$$SWF = U_1 + U_2 + \cdots + U_N$$

Equal weight on every util. With diminishing marginal utility of income, favors *some* redistribution from rich to poor.

Rawlsian

$$SWF = \min(U_1, U_2, \ldots, U_N)$$

Maximize the well-being of the *worst-off* member — generally implies *more* redistribution than utilitarianism.

Other Social Justice Principles

1. **Commodity egalitarianism**: society should guarantee basic needs (education, health care, retirement) as rights; beyond that, distribution doesn't matter.
2. **Equality of opportunity**: individuals should be compensated for circumstances beyond their control (family background, ability) — but not for the consequences of their own choices.
3. **Just deserts**: compensation should match contribution — taxes should track the benefits an individual receives from government (a libertarian view).

None of these reduce cleanly to utilitarian or Rawlsian social welfare — real policy debates mix all three.

What Do People Actually Think Is Fair?

Saez & Stantcheva (2016) surveyed ≈1,100 US respondents with hypothetical vignettes: two people have identical pre-tax and post-tax income — who deserves a €1,000 tax break more?

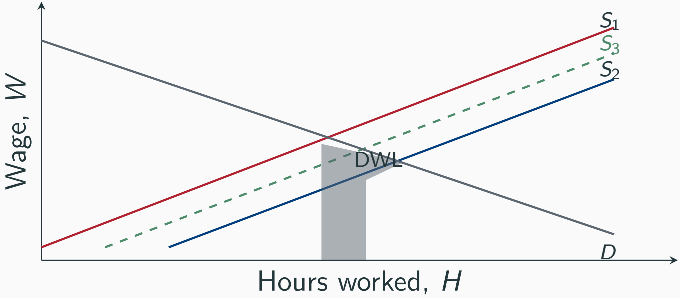
- **Consumption-lover vs. frugal person**: 74.4% say equally deserving; only 4.1% favor the consumption-lover (against pure utilitarianism, which would favor the higher-MU consumption-lover).
- **Hard-working (long hours, low wage) vs. leisure-lover (short hours, high wage)**: 42.7% favor the hard-working person; only 2.9% favor the leisure-lover.
- **Ranking deservingness of a benefit increase**: disabled person ranked most deserving 57.5% of the time; a welfare recipient *not* looking for work ranked *least* deserving 70.8% of the time.

People reward effort and penalize perceived free-riding — fairness judgments do not match any single textbook SWF.

Source: Saez, E. and Stantcheva, S. (2016), "Generalized Social Marginal Welfare Weights for Optimal Tax Theory," *American Economic Review*.

2.4 Welfare Implications of Reforming Sozialhilfe

Efficiency: The Labor Market View



S_1 no Sozialhilfe; S_2 base regime (100% withdrawal); S_3 with the reform (65% withdrawal); D = labor demand.

Sozialhilfe shifts supply left ($S_1 \to S_2$); the reform shifts it partly back ($S_2 \to S_3$), shrinking the deadweight loss.

Equity: Single-Parent Poverty in Austria

- **36%** of single-parent households were at risk of poverty in 2024 — up from just **14%** in 2014.
- **43%** were at risk of poverty *or* social exclusion (the broader EU measure).
- Single parents spend **26%** of income on housing, versus **12%** for two-parent, two-child households.

Single parents — the population Sozialhilfe most directly supports — are also the population for whom benefit cuts carry the largest equity cost.

Source: Statistik Austria, EU-SILC (2024); Sozialministerium, "Armutsgefährdung und soziale Ausgrenzung von Ein-Eltern-Haushalten."

The Trade-off in Practice

- Cutting Sozialhilfe's effective reduction rate **raises efficiency** (more labor supplied, smaller deadweight loss) — *if* the substitution effect dominates.
- It also **worsens equity** for those unable to work more (illness, caregiving, no available jobs) — exactly the single parents most at risk of poverty.
- Whichever social welfare function — or "just deserts" criterion — you find most persuasive, the trade-off does not disappear; it only shifts *where* you draw the line.

2.5 Conclusion

Highlights

- Constrained utility maximization (indifference curves + budget constraint) is the workhorse model of individual choice; profit maximization plays the same role for firms.
- Price changes split into substitution and income effects — both matter, and can point in different directions for the labor-leisure choice.
- The competitive equilibrium maximizes total surplus (1st Welfare Theorem), but says nothing about its distribution.
- The 2nd Welfare Theorem's promise of costless redistribution fails once government cannot observe the characteristics it would need to target — this *is* the equity–efficiency trade-off.
- Real people's fairness judgments (Saez–Stantcheva) don't match any single textbook social welfare function — effort and perceived deservingness matter enormously.
- Theory predicts the *direction* of Sozialhilfe reform's effects on labor supply, not their *size* — that requires the empirical tools of Lecture 3.

Discussion Questions for Next Time

1. Julia's reduction rate falls from 100% to 65%. Sketch a case (using indifference curves) where she works *less*, not more, after the reform.
2. Is Vienna's Richtwertmietzins price ceiling more like a market failure correction or a redistribution tool? Does the 1st Welfare Theorem apply?
3. Using the Saez–Stantcheva survey results, would you expect public support for Sozialhilfe to be higher or lower than support for old-age pensions? Why?
4. Pick one social justice principle (commodity egalitarianism, equality of opportunity, just deserts) and use it to argue for *or* against the 2019 Sozialhilfe reform.

Sources (1/2): Textbooks & Theory

Textbooks:

- Gruber, J. *Public Finance and Public Policy*, Ch. 2, Worth Publishers.
- Stantcheva, S. "Lecture 2: Theoretical Tools for Public Economics," Harvard EC2450A (Fall 2022).
- Becker, G. and Murphy, K. (1988), "A Rational Theory of Addiction," *Journal of Political Economy*.
- Rawls, J. *A Theory of Justice* (1971, rev. 1999), Harvard University Press.
- Saez, E. and Stantcheva, S. (2016), "Generalized Social Marginal Welfare Weights for Optimal Tax Theory," *American Economic Review*.

Sources (2/2): Austrian Data & Institutions

- Sozialhilfe-Grundsatzgesetz, BGBl. I Nr. 41/2019, § 7 Abs. 6 (earnings disregard); Sozialministerium, "Armutsgefährdung und soziale Ausgrenzung von Ein-Eltern-Haushalten"
- ÖGB, "So viel Sozialhilfe bekommt man in Österreich"
- Statistik Austria, EU-SILC (2024) income and living conditions
- Arbeiterkammer Wien, "Richtwertmietzins"; Stadt Wien, Mietenrechner

Thank you!

Questions and discussion